

UNIT 2: DERIVATIVES (Formulasheet)

Algebra of derivatives

$$1) \frac{d}{dx}(ku) = k \frac{d}{dx}(u)$$

$$2) \frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$$

$$3) \frac{d}{dx}(u.v) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$4) \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

General Formulae

$$1) \frac{d}{dx}(x^n) = n.x^{n-1}$$

$$2) \frac{d}{dx}(k) = 0$$

$$3) \frac{d}{dx}(\sin x) = \cos x$$

$$4) \frac{d}{dx}(\tan x) = \sec^2 x$$

$$5) \frac{d}{dx}(\sec x) = \sec x . \tan x$$

$$6) \frac{d}{dx}(\cos x) = -\sin x$$

$$7) \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$8) \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x . \cot x$$

$$9) \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$10) \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$11) \frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$12) \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$13) \frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$14) \frac{d}{dx}(\operatorname{cosec}^{-1} x) = -\frac{1}{x\sqrt{x^2-1}}$$

$$15) \frac{d}{dx}(e^x) = e^x$$

$$16) \frac{d}{dx}(a^x) = a^x \log a$$

$$17) \frac{d}{dx}(\log x) = \frac{1}{x}$$

$$18) \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$19) \frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

$$20) \frac{d}{dx}(x) = 1$$

Composite Function or Chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

<u>Expression</u>	<u>Substitution</u>
$\sqrt{a^2 - x^2}$	$x = a \sin \theta \text{ or } a \cos \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta \text{ or } a \cot \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta \text{ or } a \operatorname{cosec} \theta$
$\sqrt{a-x} \text{ and } \sqrt{a+x}$	$x = a \cos \theta \text{ or } a \cos 2\theta$
$1 - 2x^2$	$x = \sin \theta$
$2x^2 - 1$	$x = \cos \theta$
$\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}, \frac{2x}{1-x^2}$	$x = \tan \theta$

Logarithmic Functions

A function in the form $y = f(x)^{g(x)}$ cannot be differentiated directly because it is different from $y = f(x)^n$. In such cases we take logarithm on both sides, simplify the R.H.S and then differentiate both sides w.r.t x and obtain $\frac{dy}{dx}$. Such type of differentiation is called logarithmic differentiation.

Implicit Functions

If a variable y can be expressed as a function of another variable x i.e $y = f(x)$ then the function $f(x)$ is called the **explicit function** of x

$$eg: y = x^2 + x - 3, \quad y = \sin 3x + e^x$$

If neither x nor y can be expressed explicitly as a function of other variable then such a function is called **implicit function**.

$$eg: \log(x + y) = \sin(x + y), \quad x^m y^n = (x + y)^{m+n}$$

- While taking the derivative of implicit function, function of y is considered as a composite function of x .

Parametric Functions

$$\text{If } x = f(t); y = g(t) \text{ , then } \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Second Order Derivatives

Let $y = f(x)$. Then

$$\frac{dy}{dx} = f'(x). \quad \dots\dots\dots(1)$$

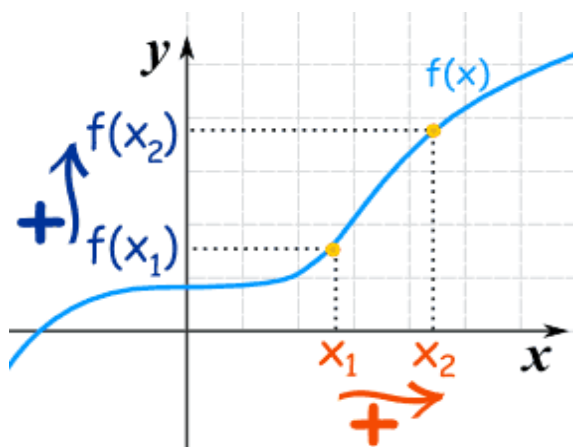
If $f'(x)$ is differentiable, we may differentiate it again with respect to x . Then the L.H.S of (1) becomes $\frac{d}{dx} \left(\frac{dy}{dx} \right)$ which is called the second order derivative of y with respect to x and it is denoted by $\frac{d^2 y}{dx^2}$. The second order derivative of $f(x)$ is also denoted by $f''(x)$ or $D^2 y$ or y_2 , if $y = f(x)$.

The higher order derivatives may be defined similarly.

Rate of change of quantities

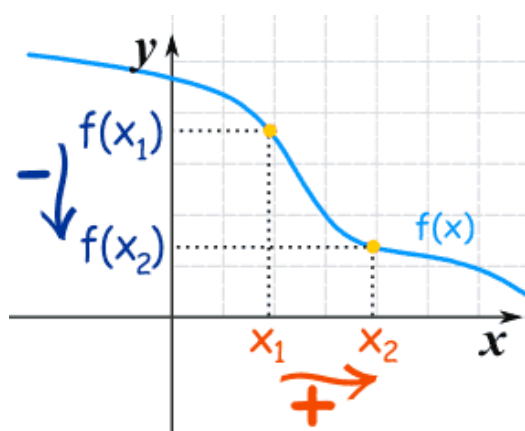
Whenever one quantity y varies with another quantity x , satisfying some rule $y = f(x)$, then $\frac{dy}{dx} = f'(x)$ represents the rate of change of y with respect to x and $\left. \frac{dy}{dx} \right|_{x=x_0}$ or $f'(x_0)$ represents the rate of change of y with respect to x at $x = x_0$.

Increasing and Decreasing Function



Def: A function $y = f(x)$ is said to be an **increasing function** on an interval, if $x_1 < x_2$ in interval implies that $f(x_1) \leq f(x_2)$.

Def: A function $y = f(x)$ is said to be an **strictly increasing function** on an interval, if $x_1 < x_2$ in interval implies that $f(x_1) < f(x_2)$.



Def: A function $y = f(x)$ is said to be a **decreasing function** on an interval, if $x_1 < x_2$ in interval implies that $f(x_1) \geq f(x_2)$.

Def: A function $y = f(x)$ is said to be a **strictly decreasing function** on an interval, if $x_1 < x_2$ in interval implies that $f(x_1) > f(x_2)$.

Theorem: Let f be continuous on $[a, b]$ and differentiable on the open interval (a, b) . Then

i) f is strictly increasing in $[a, b]$ if $f'(x) > 0$ for each $x \in (a, b)$.

ii) f is strictly decreasing in $[a, b]$ if $f'(x) < 0$ for each $x \in (a, b)$.

iii) f is a constant function in $[a, b]$ if $f'(x) = 0$ for each $x \in (a, b)$.

Maxima and Minima(Second Derivative Test)

Let f be a function defined on an interval I and $c \in I$. Let f be twice differentiable at c . Then

i) $x = c$ is a point of local maxima if $f'(c) = 0$ and $f''(c) < 0$. Here $f(c)$ is maximum value of f .

ii) $x = c$ is a point of local minima if $f'(c) = 0$ and $f''(c) > 0$. Here $f(c)$ is minimum value of f .

iii) The test fails if $f'(c) = 0$ and $f''(c) = 0$